Intro to Machine Learning
Quiz 3 Practice Problems
Practice problems, not turned in

Directions: These practice problems cover the material for Quiz 3: linear discriminant analysis in Part 1, quadratic discriminant analysis in Part 2, and basis functions and splines in Part 3. They are not turned in, but the quiz draws on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording.

Part 1: Linear Discriminant Analysis

Notation follows ISL 4.4. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. When fitting LDA by hand, use the pooled variance with divisor $n - K$.

1. **The LDA discriminant.** (ISL 4.8, Exercise 2; ISL 4.4.1–4.4.2, ESL 4.3.)

   (a) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma^2)$, with the same σ^2 in every class. Show that the Bayes classifier assigns x to the class with the largest

   $$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

   (b) With $K = 3$ and $p = 4$, how many parameters does LDA estimate (priors, means and the shared covariance)? How many coefficients does multinomial logistic regression estimate? The LDA posterior has the same form as a multinomial logistic regression, so what is LDA doing with the extra parameters?

2. **LDA with one predictor.** (ISL 4.8, Exercise 7; ISL 4.4.2.) We want to predict whether a stock will issue a dividend this year (yes or no) from X, last year's percentage profit. Among companies that issued a dividend, X averaged $\bar{x} = 10$; among those that didn't, $\bar{x} = 0$. The variance of X in both groups is $\hat{\sigma}^2 = 36$, and 80% of companies issued dividends. Assume X is normal in each group. Find the probability that a company with $X = 4$ issues a dividend this year.

3. **Fitting LDA by hand.** (ISL 4.4.2.) A training set has one predictor:

   $$\text{class 1: } x = 1, 2, 3; \qquad \text{class 2: } x = 5, 6, 6, 7.$$

   (a) Compute $\hat{\pi}_k$, $\hat{\mu}_k$ and the pooled variance $\hat{\sigma}^2$.

   (b) Find the LDA decision boundary. Which way did the unequal priors move it from the midpoint of the means?

Part 2: Quadratic Discriminant Analysis

QDA assumes $X \mid Y = k \sim N(\mu_k, \Sigma_k)$; naive Bayes assumes $f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$.

4. **QDA is quadratic.** (ISL 4.8, Exercise 3; ISL 4.4.3, ESL 4.3.) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$, with a different variance in each class. Find the discriminant $\delta_k(x)$ the Bayes classifier maximizes.

5. **QDA by hand: same means, different spreads.** (ISL 4.4.3.) Two equally likely classes have $X \mid Y = 1 \sim N(0, 1)$ and $X \mid Y = 2 \sim N(0, 4)$.

   (a) Find the Bayes decision rule. Where is each class predicted?

   (b) Now let $\pi_1 = 0.8$. How does the rule change?

6. **LDA or QDA?** (ISL 4.8, Exercise 5; ISL 4.4.3, 4.5.) Answer each and explain.

   (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (c) In general, as n increases, do we expect the test accuracy of QDA relative to LDA to improve, decline or stay the same?

   (d) True or false: even if the Bayes boundary is linear, QDA will probably achieve a lower test error rate than LDA, because QDA is flexible enough to model a linear boundary.

   (e) With $p = 10$ predictors and $K = 3$ classes, how many covariance parameters do LDA, QDA and Gaussian naive Bayes each estimate? How many observations does each class need for QDA's $\hat{\Sigma}_k$ to be invertible?

Part 3: Basis Functions and Splines

Here $I(\cdot)$ is the indicator, equal to 1 when its condition holds and 0 otherwise.

7. **Degrees of freedom and variance.** (ISL 7.3, 7.4.4; ESL 7.3; the lecture's pointwise-variance and polynomials-versus-splines sections.) A model with basis functions $b_1, \ldots, b_J$ and an intercept is fitted by least squares to n points, so the basis matrix $\mathbf{B}$ is $n \times (J + 1)$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{B}(\mathbf{B}^T\mathbf{B})^{-1}\mathbf{B}^T$, and the degrees of freedom are $\mathrm{tr}(\mathbf{H}) = J + 1$. Treat the x_i as fixed, with $\mathrm{Var}(\varepsilon_i) = \sigma^2$.

   (a) Give J for a degree-7 polynomial, a cubic spline with 5 knots, and a natural cubic spline with 5 interior knots.

   (b) Using $\mathrm{Var}\left(\hat{f}(x_i)\right) = \sigma^2 h_{ii}$, show that the average variance of the fit over the training points is $\sigma^2(J + 1)/n$. With $n = 100$ and $\sigma^2 = 1$, what is it for $J = 3$, 7 and 19? Does the answer depend on which basis is used?

   (c) A cubic spline with 4 interior knots and a natural cubic spline with 6 interior knots both have $J = 7$. Compare their total variance over the training points and where along x each puts it. Which would you use if f wiggles a lot in the middle of the data but is close to linear near the ends?

8. **Sketching a fit from its basis functions, I.** (ISL 7.9, Exercise 3; ISL 7.3.) A curve is fitted with basis functions $b_1(X) = X$ and $b_2(X) = (X-1)^2 I(X \geq 1)$. The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = -2$. Sketch the fitted curve for $-2 \leq X \leq 2$, marking the intercepts, slopes and anything else of interest.

9. **Sketching a fit from its basis functions, II.** (ISL 7.9, Exercise 4; ISL 7.2, 7.3.) A curve is fitted with basis functions

$$b_1(X) = I(0 \leq X \leq 2) - (X-1) I(1 \leq X \leq 2),$$

$$b_2(X) = (X-3) I(3 \leq X \leq 4) + I(4 < X \leq 5).$$

The regression $Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + \varepsilon$ gives $\hat{\beta}_0 = 1$, $\hat{\beta}_1 = 1$ and $\hat{\beta}_2 = 3$. Sketch the fitted curve for $-2 \leq X \leq 6$, marking the intercepts, slopes and anything else of interest.

10. **Exploring df with the basis expansions app.** (ISL 7.4.4–7.4.5; Problem 7.) Open the Basis Expansions app and press **Reset**, which gives the ESL wave $f(x) = \sin(12(x+0.2))/(x+0.2)$ with $\sigma = 1$ and $n = 100$. The bottom panel plots training MSE, LOOCV and test error against df for the selected basis; the app's df is $J + 1$ (the intercept counts). After Reset the sample is always the same, so your numbers should match the solutions.

    (a) With the **Polynomial** basis, move the degree from 1 to 15. Describe what happens to the training MSE, the test error and the $\pm 2\,\mathrm{SE}$ band. At what df is the test error smallest? What is the lowest the test error could ever be, and why?

    (b) Switch to the **Natural cubic spline** and move K from 0 to 15. At what df is the test error smallest? Compare the test error at the largest df with the polynomial's at its largest df. Which basis is hurt less by too much flexibility, and why?

    (c) For the natural spline, find two neighbouring values of K where the training MSE goes *up* as K increases. Why can that happen here, when the polynomial's training MSE never goes up?

    (d) Does LOOCV pick roughly the same df as the test error, for both bases? Which of the two curves could you compute on a real data set?

    (e) Keep the natural spline and set $n = 30$, then $n = 500$. How does the df with the smallest test error change? Explain using Problem 7. What happens to the polynomial at $n = 30$ once d reaches 8?